



August 2016

## 2N7000 / 2N7002 / NDS7002A

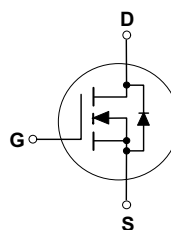
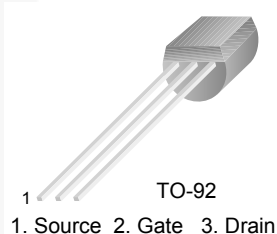
### N-Channel Enhancement Mode Field Effect Transistor

#### Features

- High Density Cell Design for Low  $R_{DS(ON)}$
- Voltage Controlled Small Signal Switch
- Rugged and Reliable
- High Saturation Current Capability

#### Description

These N-channel enhancement mode field effect transistors are produced using Fairchild's proprietary, high cell density, DMOS technology. These products have been designed to minimize on-state resistance while providing rugged, reliable, and fast switching performance. They can be used in most applications requiring up to 400 mA DC and can deliver pulsed currents up to 2 A. These products are particularly suited for low-voltage, low-current applications, such as small servo motor control, power MOSFET gate drivers, and other switching applications.



#### Ordering Information

| Part Number | Marking | Package   | Packing Method | Min Order Qty / Immediate Pack Qty |
|-------------|---------|-----------|----------------|------------------------------------|
| 2N7000      | 2N7000  | TO-92 3L  | Bulk           | 10000 / 1000                       |
| 2N7000_D74Z | 2N7000  | TO-92 3L  | Ammo           | 2000 / 2000                        |
| 2N7000_D75Z | 2N7000  | TO-92 3L  | Tape and Reel  | 2000 / 2000                        |
| 2N7000_D26Z | 2N7000  | TO-92 3L  | Tape and Reel  | 2000 / 2000                        |
| 2N7002      | 702     | SOT-23 3L | Tape and Reel  | 3000 / 3000                        |
| NDS7002A    | 712     | SOT-23 3L | Tape and Reel  | 3000 / 3000                        |

## Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at  $T_C = 25^\circ\text{C}$  unless otherwise noted.

| Symbol                            | Parameter                                                                           | Value      |        |            | Unit  |
|-----------------------------------|-------------------------------------------------------------------------------------|------------|--------|------------|-------|
|                                   |                                                                                     | 2N7000     | 2N7002 | NDS7002A   |       |
| V <sub>DSS</sub>                  | Drain-to-Source Voltage                                                             | 60         |        |            | V     |
| V <sub>DGR</sub>                  | Drain-Gate Voltage (R <sub>GS</sub> ≤ 1 MΩ )                                        | 60         |        |            | V     |
| V <sub>GSS</sub>                  | Gate-Source Voltage - Continuous                                                    | ±20        |        |            | V     |
|                                   | Gate-Source Voltage - Non Repetitive (tp < 50 μS)                                   | ±40        |        |            |       |
| I <sub>D</sub>                    | Maximum Drain Current - Continuous                                                  | 200        | 115    | 280        | mA    |
|                                   | Maximum Drain Current - Pulsed                                                      | 500        | 800    | 1500       |       |
| P <sub>D</sub>                    | Maximum Power Dissipation Derated above 25°C                                        | 400        | 200    | 300        | mW    |
|                                   |                                                                                     | 3.2        | 1.6    | 2.4        | mW/°C |
| T <sub>J</sub> , T <sub>STG</sub> | Operating and Storage Temperature Range                                             | -55 to 150 |        | -65 to 150 | °C    |
| T <sub>L</sub>                    | Maximum Lead Temperature for Soldering Purposes, 1/16-inch from Case for 10 Seconds | 300        |        |            | °C    |

## Thermal Characteristics

Values are at  $T_C = 25^\circ\text{C}$  unless otherwise noted.

| Symbol          | Parameter                               | Value  |        |          | Unit                      |
|-----------------|-----------------------------------------|--------|--------|----------|---------------------------|
|                 |                                         | 2N7000 | 2N7002 | NDS7002A |                           |
| $R_{\theta JA}$ | Thermal Resistance, Junction to Ambient | 312.5  | 625    | 417      | $^\circ\text{C}/\text{W}$ |

## Electrical Characteristics

Values are at  $T_C = 25^\circ\text{C}$  unless otherwise noted.

| Symbol                     | Parameter                       | Conditions                                                           | Type               | Min. | Typ. | Max. | Unit          |
|----------------------------|---------------------------------|----------------------------------------------------------------------|--------------------|------|------|------|---------------|
| <b>Off Characteristics</b> |                                 |                                                                      |                    |      |      |      |               |
| $BV_{DSS}$                 | Drain-Source Breakdown Voltage  | $V_{GS} = 0\text{ V}, I_D = 10\text{ }\mu\text{A}$                   | All                | 60   |      |      | V             |
| $I_{DSS}$                  | Zero Gate Voltage Drain Current | $V_{DS} = 48\text{ V}, V_{GS} = 0\text{ V}$                          | 2N7000             |      |      | 1    | $\mu\text{A}$ |
|                            |                                 | $V_{DS} = 48\text{ V}, V_{GS} = 0\text{ V}, T_C = 125^\circ\text{C}$ |                    |      |      | 1    | mA            |
|                            |                                 | $V_{DS} = 60\text{ V}, V_{GS} = 0\text{ V}$                          | 2N7002             |      |      | 1    | $\mu\text{A}$ |
|                            |                                 | $V_{DS} = 60\text{ V}, V_{GS} = 0\text{ V}, T_C = 125^\circ\text{C}$ | NDS7002A           |      |      | 0.5  | mA            |
| $I_{GSSF}$                 | Gate - Body Leakage, Forward    | $V_{GS} = 15\text{ V}, V_{DS} = 0\text{ V}$                          | 2N7000             |      |      | 10   | nA            |
|                            |                                 | $V_{GS} = 20\text{ V}, V_{DS} = 0\text{ V}$                          | 2N7002<br>NDS7002A |      |      | 100  | nA            |
| $I_{GSSR}$                 | Gate - Body Leakage, Reverse    | $V_{GS} = -15\text{ V}, V_{DS} = 0\text{ V}$                         | 2N7000             |      |      | -10  | nA            |
|                            |                                 | $V_{GS} = -20\text{ V}, V_{DS} = 0\text{ V}$                         | 2N7002<br>NDS7002A |      |      | -100 | nA            |

**Electrical Characteristics** (Continued)

| Symbol                    | Parameter                         | Conditions                                                             | Type               | Min. | Typ. | Max. | Unit     |
|---------------------------|-----------------------------------|------------------------------------------------------------------------|--------------------|------|------|------|----------|
| <b>On Characteristics</b> |                                   |                                                                        |                    |      |      |      |          |
| $V_{GS(th)}$              | Gate Threshold Voltage            | $V_{DS} = V_{GS}, I_D = 1 \text{ mA}$                                  | 2N7000             | 0.8  | 2.1  | 3    | V        |
|                           |                                   | $V_{DS} = V_{GS}, I_D = 250 \mu\text{A}$                               | 2N7002<br>NDS7002A | 1    | 2.1  | 2.5  |          |
| $R_{DS(on)}$              | Static Drain-Source On-Resistance | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | 2N7000             |      | 1.2  | 5    | $\Omega$ |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}, T_C = 125^\circ\text{C}$ |                    |      | 1.9  | 9    |          |
|                           |                                   | $V_{GS} = 4.5 \text{ V}, I_D = 75 \text{ mA}$                          |                    |      | 1.8  | 5.3  |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | 2N7002             |      | 1.2  | 7.5  |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}, T_C = 100^\circ\text{C}$ |                    |      | 1.7  | 13.5 |          |
|                           |                                   | $V_{GS} = 5 \text{ V}, I_D = 50 \text{ mA}$                            |                    |      | 1.7  | 7.5  |          |
|                           |                                   | $V_{GS} = 5 \text{ V}, I_D = 50 \text{ mA}, T_C = 100^\circ\text{C}$   |                    |      | 2.4  | 13.5 |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | NDS7002A           |      | 1.2  | 2    |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}, T_C = 125^\circ\text{C}$ |                    |      | 2    | 3.5  |          |
|                           |                                   | $V_{GS} = 5 \text{ V}, I_D = 50 \text{ mA}$                            |                    |      | 1.7  | 3    |          |
|                           |                                   | $V_{GS} = 5 \text{ V}, I_D = 50 \text{ mA}, T_C = 125^\circ\text{C}$   |                    |      | 2.8  | 5    |          |
| $V_{DS(on)}$              | Drain-Source On-Voltage           | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | 2N7000             |      | 0.6  | 2.5  | V        |
|                           |                                   | $V_{GS} = 4.5 \text{ V}, I_D = 75 \text{ mA}$                          |                    |      | 0.14 | 0.4  |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | 2N7002             |      | 0.6  | 3.75 |          |
|                           |                                   | $V_{GS} = 5.0 \text{ V}, I_D = 50 \text{ mA}$                          |                    |      | 0.09 | 1.5  |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, I_D = 500 \text{ mA}$                          | NDS7002A           |      | 0.6  | 1    |          |
|                           |                                   | $V_{GS} = 5.0 \text{ V}, I_D = 50 \text{ mA}$                          |                    |      | 0.09 | 0.15 |          |
| $I_{D(on)}$               | On-State Drain Current            | $V_{GS} = 4.5 \text{ V}, V_{DS} = 10 \text{ V}$                        | 2N7000             | 75   | 600  |      | mA       |
|                           |                                   | $V_{GS} = 10 \text{ V}, V_{DS} \geq 2 V_{DS(on)}$                      | 2N7002             | 500  | 2700 |      |          |
|                           |                                   | $V_{GS} = 10 \text{ V}, V_{DS} \geq 2 V_{DS(on)}$                      | NDS7002A           | 500  | 2700 |      |          |
| $g_{FS}$                  | Forward Transconductance          | $V_{DS} = 10 \text{ V}, I_D = 200 \text{ mA}$                          | 2N7000             | 100  | 320  |      | mS       |
|                           |                                   | $V_{DS} \geq 2 V_{DS(on)}, I_D = 200 \text{ mA}$                       | 2N7002             | 80   | 320  |      |          |
|                           |                                   | $V_{DS} \geq 2 V_{DS(on)}, I_D = 200 \text{ mA}$                       | NDS7002A           | 80   | 320  |      |          |

**Electrical Characteristics** (Continued)

| Symbol                                                 | Parameter                                             | Conditions                                                                                                                     | Type               | Min. | Typ. | Max. | Unit |
|--------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------|------|------|------|------|
| Dynamic Characteristics                                |                                                       |                                                                                                                                |                    |      |      |      |      |
| C <sub>iss</sub>                                       | Input Capacitance                                     | V <sub>DS</sub> = 25 V, V <sub>GS</sub> = 0 V,<br>f = 1.0 MHz                                                                  | All                |      | 20   | 50   | pF   |
| C <sub>oss</sub>                                       | Output Capacitance                                    |                                                                                                                                | All                |      | 11   | 25   |      |
| C <sub>rss</sub>                                       | Reverse Transfer Capacitance                          |                                                                                                                                | All                |      | 4    | 5    |      |
| t <sub>on</sub>                                        | Turn-On Time                                          | V <sub>DD</sub> = 15 V, R <sub>L</sub> = 25 Ω,<br>I <sub>D</sub> = 500 mA,<br>V <sub>GS</sub> = 10 V, R <sub>GEN</sub> = 25 Ω  | 2N7000             |      |      | 10   | ns   |
|                                                        |                                                       | V <sub>DD</sub> = 30 V, R <sub>L</sub> = 150 Ω,<br>I <sub>D</sub> = 200 mA, V <sub>GS</sub> = 10 V,<br>R <sub>GEN</sub> = 25 Ω | 2N7002<br>NDS7002A |      |      | 20   |      |
| t <sub>off</sub>                                       | Turn-Off Time                                         | V <sub>DD</sub> = 15 V, R <sub>L</sub> = 25 Ω ,<br>I <sub>D</sub> = 500 mA, V <sub>GS</sub> = 10 V,<br>R <sub>GEN</sub> = 25 Ω | 2N7000             |      |      | 10   | ns   |
|                                                        |                                                       | V <sub>DD</sub> = 30 V, R <sub>L</sub> = 150 Ω,<br>I <sub>D</sub> = 200 mA, V <sub>GS</sub> = 10 V,<br>R <sub>GEN</sub> = 25 Ω | 2N7002<br>NDS7002A |      |      | 20   |      |
| Drain-Source Diode Characteristics and Maximum Ratings |                                                       |                                                                                                                                |                    |      |      |      |      |
| I <sub>S</sub>                                         | Maximum Continuous Drain-Source Diode Forward Current |                                                                                                                                | 2N7002             |      |      | 115  | mA   |
|                                                        |                                                       |                                                                                                                                | NDS7002A           |      |      | 280  |      |
| I <sub>SM</sub>                                        | Maximum Pulsed Drain-Source Diode Forward Current     |                                                                                                                                | 2N7002             |      |      | 0.8  | A    |
|                                                        |                                                       |                                                                                                                                | NDS7002A           |      |      | 1.5  |      |
| V <sub>SD</sub>                                        | Drain-Source Diode Forward Voltage                    | V <sub>GS</sub> = 0 V,<br>I <sub>S</sub> = 115 mA <sup>(1)</sup>                                                               | 2N7002             |      | 0.88 | 1.5  | V    |
|                                                        |                                                       | V <sub>GS</sub> = 0 V,<br>I <sub>S</sub> = 400 mA <sup>(1)</sup>                                                               | NDS7002A           |      | 0.88 | 1.2  |      |

**Note:**

1. Pulse test : Pulse Width  $\leq 300\ \mu\text{s}$ , Duty Cycle  $\leq 2\ \%$ .

# Typical Performance Characteristics

2N7000 / 2N7002 / NDS7002A

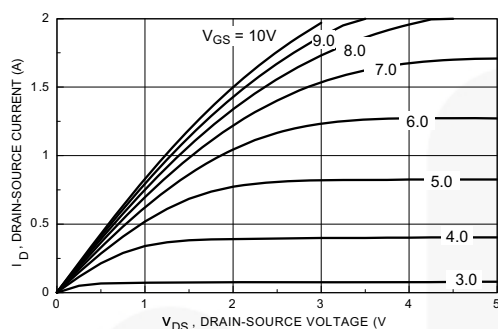


Figure 1. On-Region Characteristics

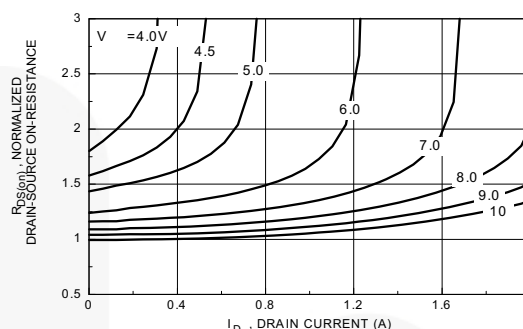


Figure 2. On-Resistance Variation with Gate Voltage and Drain Current

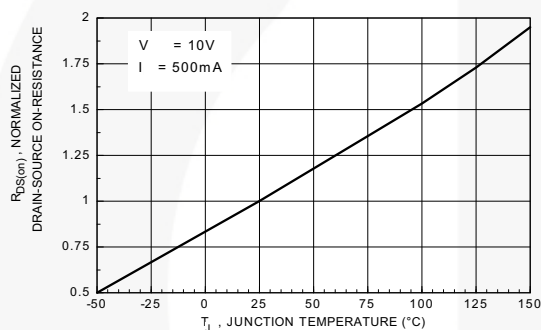


Figure 3. On-Resistance Variation with Temperature

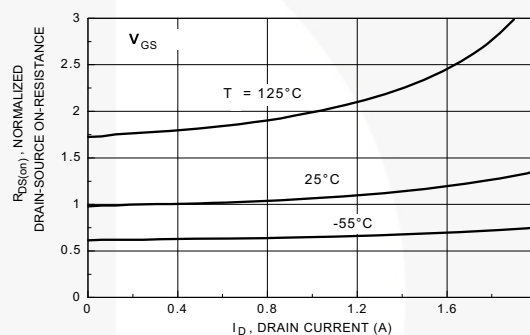


Figure 4. On-Resistance Variation with Drain Current and Temperature

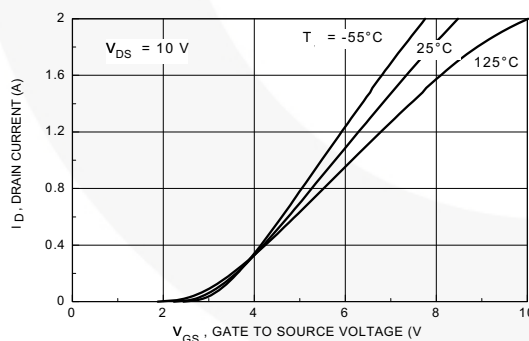


Figure 5. Transfer Characteristics

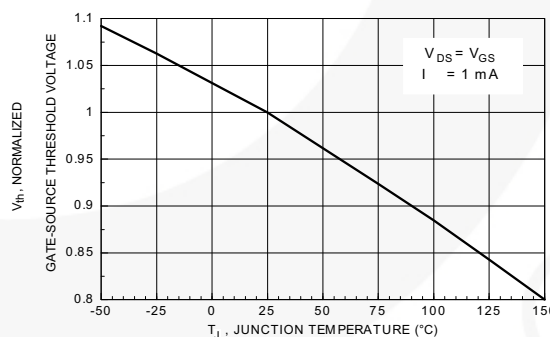


Figure 6. Gate Threshold Variation with Temperature

## Typical Performance Characteristics (Continued)

2N7000 / 2N7002 / NDS7002A

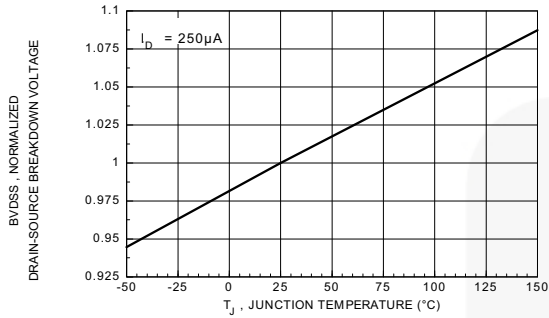


Figure 7. Breakdown Voltage Variation with Temperature

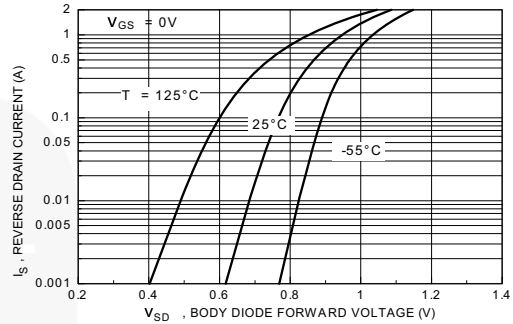


Figure 8. Body Diode Forward Voltage Variation with Temperature

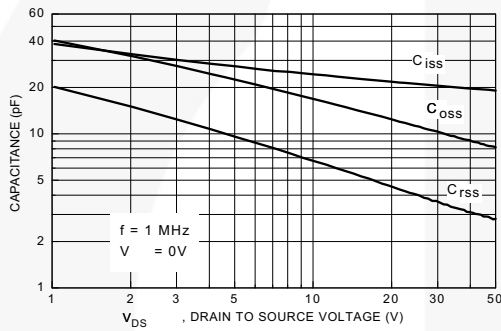


Figure 9. Capacitance Characteristics

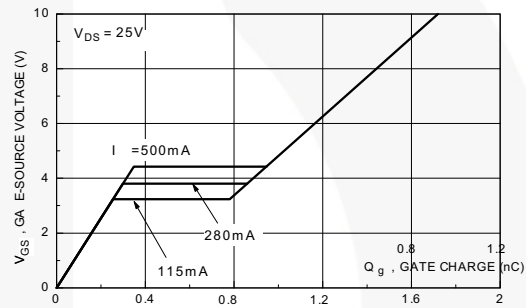


Figure 10. Gate Charge Characteristics

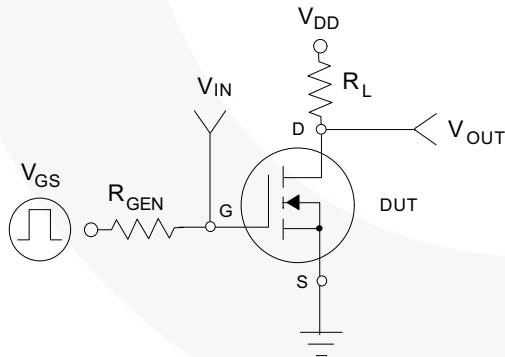


Figure 11. Switching Test Circuit

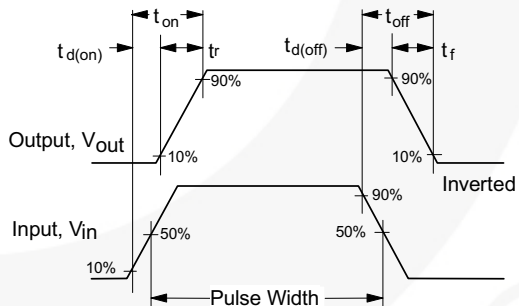


Figure 12. Switching Waveforms

## Typical Performance Characteristics (Continued)

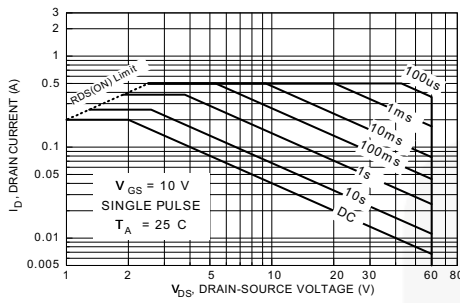


Figure 13. 2N7000 Maximum Safe Operating Area

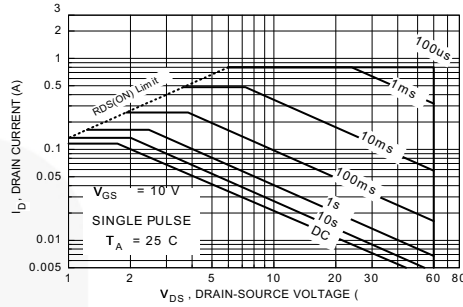


Figure 14. 2N7002 Maximum Safe Operating Area

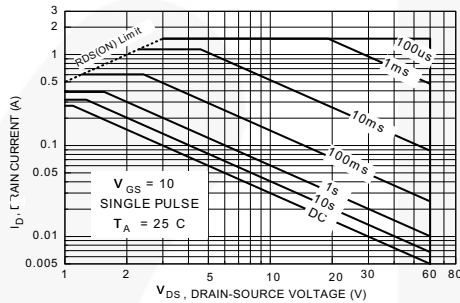


Figure 15. NDS7000A Maximum Safe Operating Area

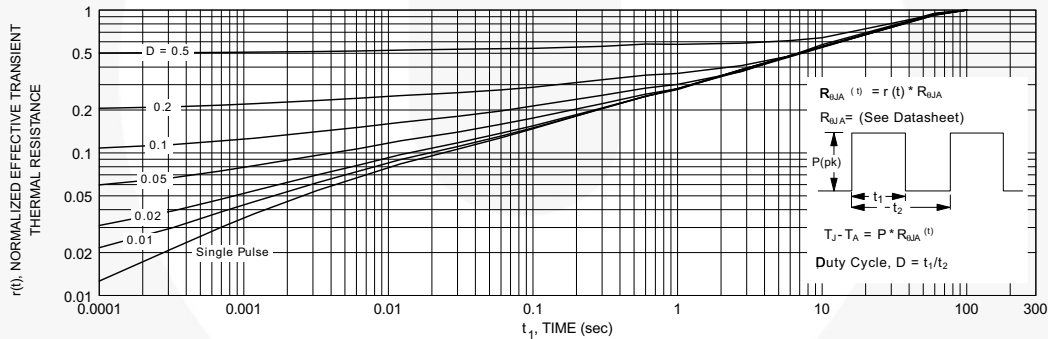


Figure 16. TO-92, 2N7000 Transient Thermal Response Curve

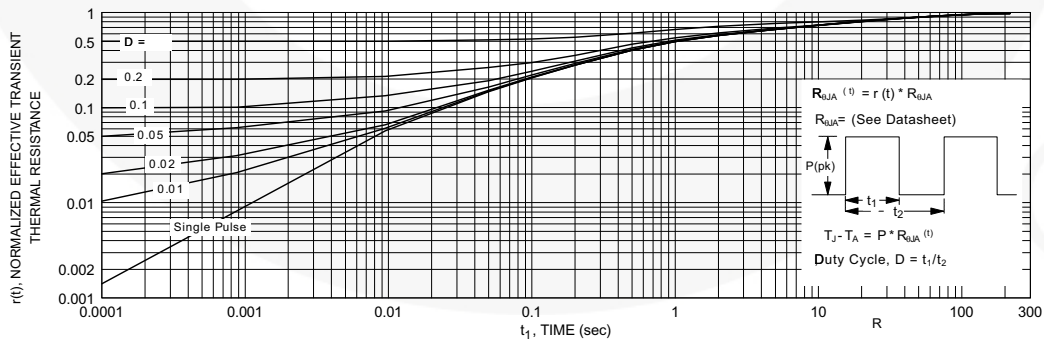
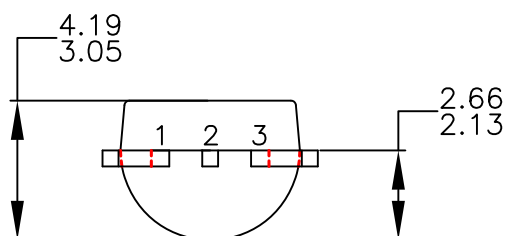
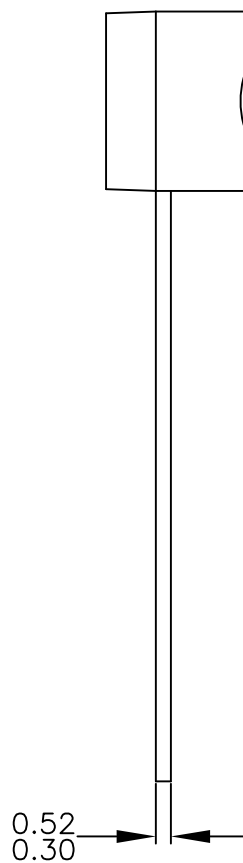
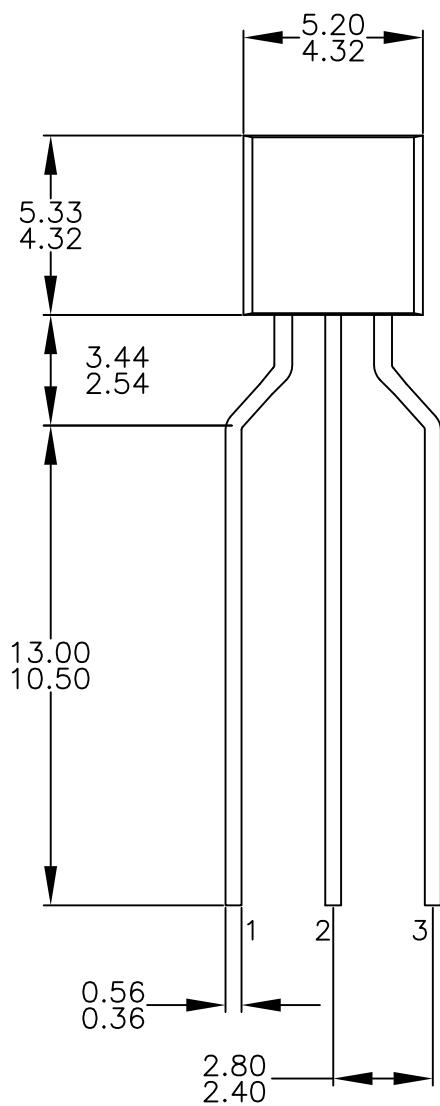


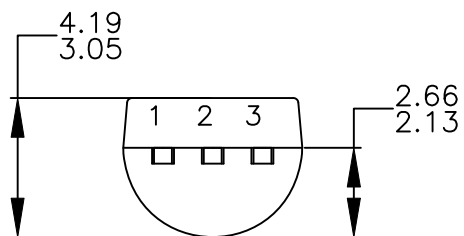
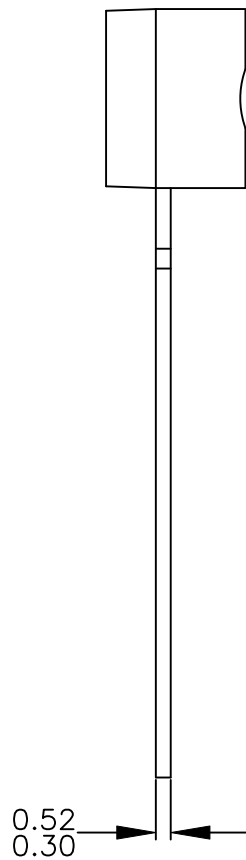
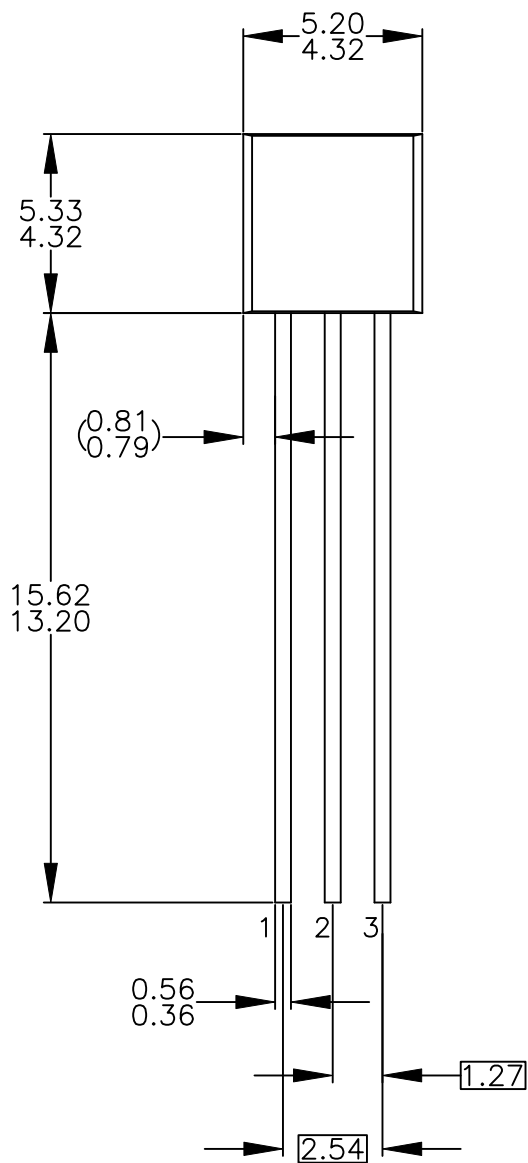
Figure 17. SOT-23, 2N7002 / NDS7002A Transient Thermal Response Curve



NOTES: UNLESS OTHERWISE SPECIFIED

- DRAWING CONFORMS TO JEDEC MS-013, VARIATION AC.
- ALL DIMENSIONS ARE IN MILLIMETERS.
- DRAWING CONFORMS TO ASME Y14.5M-2009.
- DRAWING FILENAME: MKT-ZA03FREV3.
- FAIRCHILD SEMICONDUCTOR.

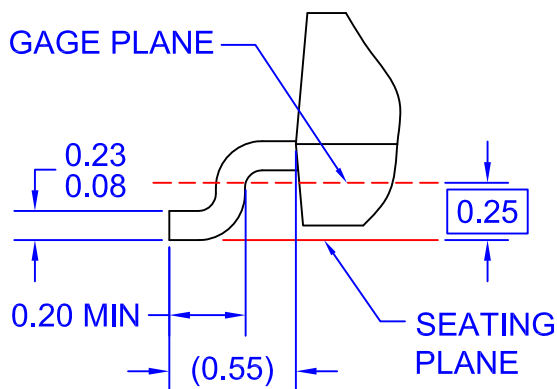
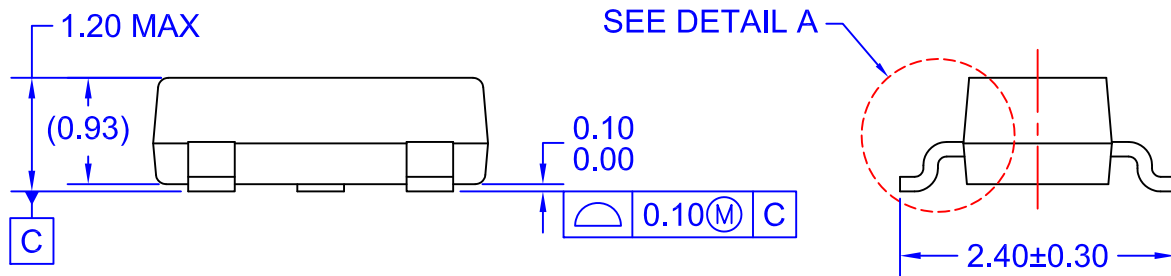
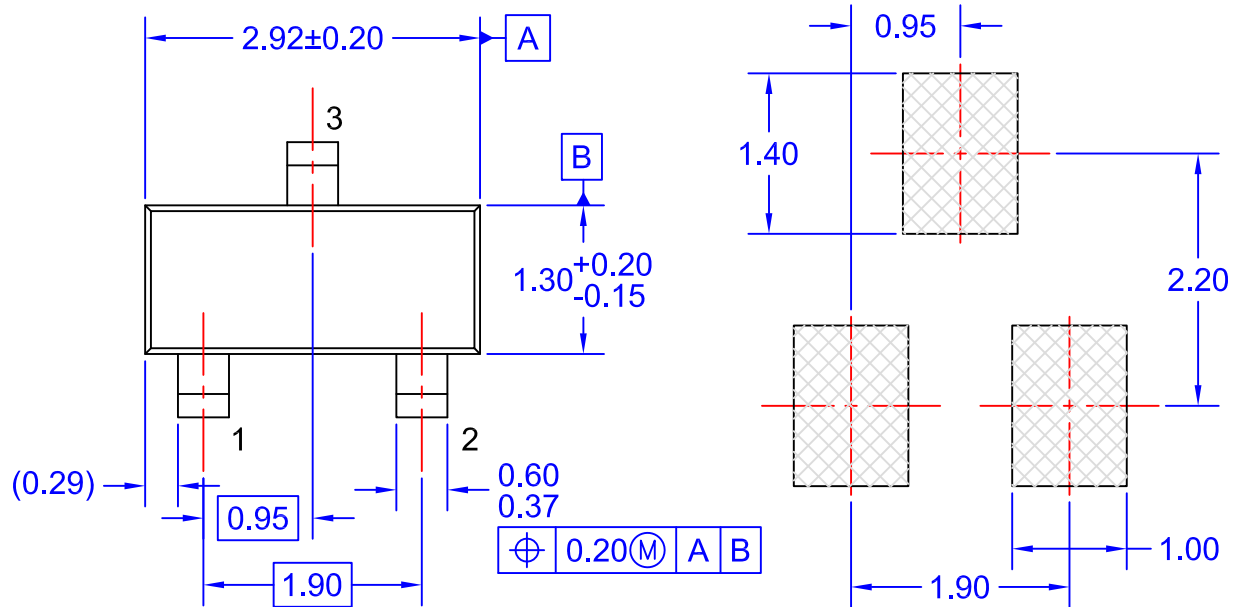




NOTES: UNLESS OTHERWISE SPECIFIED

- A) DRAWING WITH REFERENCE TO JEDEC TO-92 RECOMMENDATIONS.
- B) ALL DIMENSIONS ARE IN MILLIMETERS.
- C) DRAWING CONFORMS TO ASME Y14.5M-2009.
- D) DRAWING FILENAME: MKT-ZA03DREV4.





**DETAIL A**  
SCALE: 2X

NOTES: UNLESS OTHERWISE SPECIFIED

- A) REFERENCE JEDEC REGISTRATION TO-236, VARIATION AB, ISSUE H.
- B) ALL DIMENSIONS ARE IN MILLIMETERS.
- C) DIMENSIONS ARE INCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR EXTRUSIONS.
- D) DIMENSIONING AND TOLERANCING PER ASME Y14.5M - 2009.
- E) DRAWING FILE NAME: MA03DREV12





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MegaBuck™  
MICROCOUPLER™  
MicroFET™  
MicroPak™  
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MotionMax™  
MotionGrid®  
MTI®  
MTX®  
MVN®  
mWSaver®  
OptoHit™  
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SuperSOT™-6  
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